Discrete Probability Problems

Nine problems from ICPC-style contests, with solutions

These problems come from programming contests. Each one keeps the probability idea of the original and drops the programming layer (large inputs, modular output, data structures), using small concrete numbers instead. Only discrete probability is used. Each problem asks one question that rests on one key idea, and its solution names the idea first. A remark after each solution connects it to the general and contest versions; the remarks are not part of the problems. Problems 1–5 are the main set and Problems 6–9 are backups, each set in order of difficulty. The selection is explained in `prob_course_candidates.md`. Every answer was checked by computer in exact arithmetic.

Difficulty is given on the Codeforces rating scale. "Paper" is an estimate, by judgment and accurate to about ±200, of the rating a Codeforces problem would have if its whole difficulty were the math step asked here, with tiny inputs and nothing to program. Each estimate is anchored on a rated Codeforces problem with the same idea, taken from `data/cf_problemset.json`. A paper version with small numbers sits below its anchor, because the anchor also has to be coded for large inputs. "Contest" is the rating of the original contest problem from this repository's shipped DE fit (`output/problem_ratings_calibrated.json`); "—" means the problem is not in that fit. It mostly measures the programming layer: Orchard Fence is the hardest contest problem here and the easiest paper problem.

	Problem	Paper	Contest	Anchor (same idea)
1	Mingle	1300	2239	621C (1700)
2	Accidental Arithmetic	1500	1877	696B (1700)
3	Uniform Chemistry	1600	1960*	280C (2200)
4	502 Bad Gateway	1700	1972	—
5	Jinx or Jackpot	1900	2438	—
6	Orchard Fence	1100	3067	—
7	Disbursement on Quarantine Policy	1400	—	621C (1700)
8	Crashing Competition Computer	1400	2269	1265E (2100)
9	Chopsticks	1700	—	—

*Its Codeforces mirror, 1776K, is officially rated 3200; `problem_rating_audit.md` judges about 2000.

Main problems

1 Mingle

Source: NAQ 2025, Problem G (QOJ). *Topic:* indicator variables. *Difficulty:* about 1300 (CF scale).

Problem. Nine players stand in a circle, each facing one of nine rooms that are also arranged in a circle. When the music stops, each player runs into the room they face or one of its two neighbouring rooms, each with probability $\frac{1}{3}$, independently of the others. A player who ends up alone in a room wins. Find the expected number of winners.

Solution. Key idea: count rooms, not players.

Every winner is alone in a room, and a room with exactly one player holds exactly one winner. So the number of winners is the number of rooms with exactly one player:

$$W = \sum_{\text{rooms } r} X_r, \qquad X_r = \mathbf{1}[\text{room } r \text{ holds exactly one player}] .$$

Each room can be reached by exactly three players: the one facing it and the two beside that player. Each of them enters it with probability $\frac{1}{3}$, independently, so

$$\mathbb{P}(X_r = 1) = 3 \cdot \frac{1}{3} \cdot \left(\frac{2}{3}\right)^2 = \frac{4}{9}, \qquad \mathbb{E}[W] = 9 \cdot \frac{4}{9} = 4.$$

The X_r are dependent, but linearity of expectation does not need independence. □

Remark. With n players who each choose among $2k+1$ rooms, the same argument gives $n\left(\frac{2k}{2k+1}\right)^{2k}$, about n/e for large k. The contest asks for this with $n \le 456$, modulo a prime.

2 Accidental Arithmetic

Source: NWERC 2025, Problem A (QOJ). *Topic:* symmetry. *Difficulty:* about 1500 (CF scale).

Problem. You type 12345 into a broken calculator. After each of the first four digits it inserts a "+" with probability 0.45, a "−" with probability 0.45, and nothing with probability 0.1, independently. For example, it may produce $12 - 3 + 45$ or $1 + 2345$. Find the expected value of the resulting expression.

Solution. Key idea: swapping every "+" with "−" changes neither the probability of the expression nor its first term.

Write the value as $F + R$, where F is the first term (the digits before the first sign) and R is the signed sum of the other terms. Swapping every "+" and "−" gives an expression with the same probability, because both signs have probability 0.45. The swap keeps F and turns R into $-R$. So R and $-R$ have the same distribution, $\mathbb{E}[R] = 0$, and the answer is $\mathbb{E}[F]$.

The first sign comes after digit 1 with probability 0.9, after digit 2 with probability $0.1 \cdot 0.9$, and so on; with probability 0.1^4 there is no sign at all. Hence

$$\begin{aligned}\mathbb{E}[F] &= 0.9 \cdot 1 + 0.09 \cdot 12 + 0.009 \cdot 123 + 0.0009 \cdot 1234 + 0.0001 \cdot 12345 \\ &= 0.9 + 1.08 + 1.107 + 1.1106 + 1.2345 = 5.4321.\end{aligned}$$

□

Remark. For any number with prefixes $P_1, \dots, P_L = N$ and sign probability α each, the answer is $\sum_{l<L} 2\alpha(1-2\alpha)^{l-1}P_l + (1-2\alpha)^{L-1}N$. The contest allows N with up to 1000 digits. If "+" and "−" had different probabilities, the later terms would no longer cancel.

3 Uniform Chemistry

Source: SWERC 2022–2023, Problem B (QOJ). *Topic:* indicator variables. *Difficulty:* about 1600 (CF scale).

Problem. A frog sits on pad 0 of a row of pads numbered 0 to 6. Each jump takes it to a pad chosen uniformly at random among the pads ahead of it: from pad x, to one of $x+1, \dots, 6$. Find the expected number of jumps until the frog reaches pad 6.

Hint: for each pad j, find the probability that the frog lands on j.

Solution. Key idea: the jump that first carries the frog to pad j or beyond is equally likely to land on each of the pads $j, \dots, 6$.

The number of jumps equals the number of pads the frog lands on. Fix a pad j from 1 to 6. Exactly one jump takes the frog from below j to pad j or beyond. Whichever pad $x < j$ that jump starts from, it lands uniformly on $x+1, \dots, 6$; so, given that it reaches j or beyond, it is equally likely to land on each of the $7-j$ pads $j, \dots, 6$. Hence

$$\mathbb{P}(\text{the frog lands on } j) = \frac{1}{7-j} :$$

1 for pad 6, $\frac{1}{2}$ for pad 5, …, $\frac{1}{6}$ for pad 1. By linearity, the expected number of jumps is

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} = \frac{49}{20} = 2.45. \qquad \square$$

Remark. From pad a to pad n the answer is the harmonic number $1 + \frac{1}{2} + \cdots + \frac{1}{n-a}$. The landing events are in fact independent, so the number of jumps is distributed like the number of cycles of a random permutation of $n-a$ elements. The contest has several frogs (researchers racing to a new element) and asks for each one's chance to arrive first.

4 502 Bad Gateway

Source: ICPC Asia EC Online 2024 (II), Problem L (QOJ). *Topic:* geometric distribution. *Difficulty:* about 1700 (CF scale).

Problem. A timer is set to a whole number of seconds chosen uniformly from 1 to 10, and counts down to 0. Each time the timer is set, you see its value and choose: let it run down to 0, or wait one second and then reset it to a fresh uniform value from 1 to 10. Find the smallest possible expected time until the timer reaches 0.

Solution. Key idea: the number of resets is geometric.

After a reset you are exactly where you started, so a best strategy makes the same choice every time: it keeps the values in some set K and resets on the others. Say K has c values. Each draw is kept with probability $c/10$, so the number of resets is geometric with mean $\frac{10-c}{c}$, and each reset costs one second. The kept value costs itself, and on average that is the mean of K. So the expected time is

$$\frac{10-c}{c} + \operatorname{mean}(K).$$

For a given c this is smallest when $K = \{1, \dots, c\}$, with mean $\frac{c+1}{2}$. That leaves

$$g(c) = \frac{10-c}{c} + \frac{c+1}{2} = \frac{10}{c} + \frac{c-1}{2}.$$

For $c = 1, 2, \dots, 10$ this is 10, 5.5, 4.33, 4, 4, 4.17, 4.43, 4.75, 5.11, 5.5. The minimum is 4: keep the timer when it shows at most 4 (or at most 5). Never resetting costs 5.5. $\square$

Remark. For values from 1 to T, the best c is the least with $c(c+1) \geq 2T$, about $\sqrt{2T}$, and the minimum is about $\sqrt{2T} - \frac{1}{2}$ instead of $\frac{T+1}{2}$. The contest asks for the exact fraction with $T \leq 10^9$ and lets you reset at the end of any second, which does not help: if a value is worth resetting, it is worth resetting at once.

5 Jinx or Jackpot

Source: NERC 2025, Problem J (QOJ). *Topic:* conditional probability. *Difficulty:* about 1900 (CF scale).

Problem. A slot machine is either good (each round is a win with probability 0.7) or bad (probability 0.3), equally likely, and you do not know which. In each round you bet any amount up to your balance: on a win you gain the amount bet, on a loss you lose it. Even a bet of 0 shows you whether the round was a win. You start with \$1000 and play two rounds. What is the largest possible expected final balance?

Hint: decide round 2 first, after a win and after a loss.

Solution. Key idea: the expected final balance is linear in each bet, so every bet should be 0 or everything.

First, the chance of a win in round 2. Two wins have probability $\frac{1}{2}(0.7^2 + 0.3^2) = 0.29$, and a first-round win has probability 0.5. So after a win the chance is $0.29/0.5 = 0.58$; likewise, after a loss it is $\frac{1}{2}(0.3 \cdot 0.7 + 0.7 \cdot 0.3)/0.5 = 0.42$.

Round 2. With balance w and win chance q, a bet x gives expected balance $w + x(2q-1)$. After a win ($q = 0.58$) go all-in and expect $1.16\,w$; after a loss ($q = 0.42$) bet 0 and keep w.

Round 1. A bet x gives expected final balance

$$\tfrac{1}{2}(1000 + x) \cdot 1.16 + \tfrac{1}{2}(1000 - x) \cdot 1 = 1080 + 0.08\,x,$$

so bet everything. The largest expected final balance is \$1160: go all-in twice, ending with \$4000 with probability 0.29. Betting 0 in round 1 gives only \$1080.

Round 1 is a fair bet, yet staking everything on it is best: a win both doubles the money and makes the machine look good. □

Remark. For any number of rounds and any prior, induction shows that the best expected result is the balance times a factor depending only on the outcomes seen, so every bet is 0 or all-in. The contest has up to 30 rounds and up to 10^5 candidate values of the win probability.

Backup problems

6 Orchard Fence

Source: 2025 ICPC Southeast USA Regional, Division 1, Problem A; listed under Mid-Atlantic USA in `data/tagged.json` (QOJ). *Topic:* linearity of expectation. *Difficulty:* about 1100 (CF scale).

Problem. Three of the six vertices of a regular hexagon with side 1 are chosen uniformly at random. Find the expected perimeter of the triangle they form.

Solution. Key idea: the perimeter is a sum over pairs of vertices, so add up pairs instead of triangles.

The perimeter is the total length of the three pairs among the chosen vertices. Each of the 15 pairs of vertices lies in 4 of the $\binom{6}{3} = 20$ triangles, so it is a side with probability $\frac{1}{5}$. Six pairs are adjacent (length 1), six are one vertex apart (length $\sqrt{3}$) and three are opposite (length 2). By linearity,

$$\mathbb{E}[\text{perimeter}] = \frac{1}{5}\big(6 \cdot 1 + 6\sqrt{3} + 3 \cdot 2\big) = \frac{12 + 6\sqrt{3}}{5} \approx 4.478.$$ □

Remark. The contest keeps k of n trees, which are disks of diameter d; the shortest fence is the perimeter of the convex hull of their centres plus πd. For $k > 3$ the same idea works with directed pairs: $z_i \to z_j$ is a counterclockwise hull edge exactly when every other kept point lies to its left, which has probability $\binom{\ell_{ij}}{k-2}/\binom{n}{k}$, where ℓ_{ij} counts the points to the left.

7 Disbursement on Quarantine Policy

Source: ICPC Asia Taiwan 2024, Problem D (QOJ). *Topic:* complements. *Difficulty:* about 1400 (CF scale).

Problem. A train car has two rows of 50 seats, all taken, and each passenger is infected independently with probability $\frac{1}{2}$. An infected passenger is quarantined for 4 days. A healthy passenger sitting next to an infected one (directly left, right or across) is quarantined for 2 days. A healthy passenger with nobody infected next to them, but someone infected diagonally across, is quarantined for 1 day. Find the expected total number of quarantine days.

Solution. Key idea: linearity over seats; for one seat, use the complement "nobody next to me is infected".

Take a passenger with e neighbours next to them and c diagonal neighbours. The passenger, the neighbours next to them and the diagonal ones are disjoint groups, so the three events "infected", "nobody next to them infected" and "nobody diagonal infected" are independent, with probabilities $\frac{1}{2}$, 2^{-e} and 2^{-c}. The expected quarantine length is

$$4 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2}\left(1 - 2^{-e}\right) + 1 \cdot \frac{1}{2}\,2^{-e}\left(1 - 2^{-c}\right).$$

The 4 seats at the ends of the rows have $e = 2$, $c = 1$, giving $2 + \frac{3}{4} + \frac{1}{16} = \frac{45}{16}$. The other 96 seats have $e = 3$, $c = 2$, giving $2 + \frac{7}{8} + \frac{3}{64} = \frac{187}{64}$. By linearity, the expected total is

$$4 \cdot \frac{45}{16} + 96 \cdot \frac{187}{64} = \frac{45}{4} + \frac{561}{2} = \frac{1167}{4} = 291.75. \qquad \square$$

Remark. The contest grid is up to 100×100, the day counts are inputs, and some passengers are known to be infected or healthy; the same argument works with each seat's infection probability in $\{0, \frac{1}{2}, 1\}$.

8 Crashing Competition Computer

Source: NWERC 2022, Problem C (QOJ). *Topic:* recursion for expectations. *Difficulty:* about 1400 (CF scale).

Problem. You must type 3 characters, one second each. After each character, the computer crashes with probability $\frac{1}{2}$, independently, and you lose everything typed since the last save (the start counts as saved); you then continue at once. Saving takes 3 seconds and never crashes, and you must save after the last character. If you choose when to save as well as possible, what is the expected total time?

Solution. Key idea: a crash puts you back exactly where you were after the last save, so typing k characters between saves is like waiting for k heads in a row.

Let E_k be the expected time to type k characters in a row, starting from a save. Once $k-1$ are typed, you type one more; with probability $\frac{1}{2}$ it crashes and you need E_k again. So

$$E_k = E_{k-1} + 1 + \tfrac{1}{2}E_k, \qquad \text{that is,} \qquad E_k = 2E_{k-1} + 2,$$

with $E_0 = 0$. This gives $E_1 = 2$, $E_2 = 6$ and $E_3 = 14$. Since a crash only returns you to the last save, a strategy is just a choice of where to save:

- only at the end: $14 + 3 = 17$;
- after character 1, or after character 2: $(2 + 3) + (6 + 3) = 14$;
- after characters 1 and 2: $3 \cdot (2 + 3) = 15$.

The smallest expected total time is 14. □

Remark. With crash probability p and a recovery time r after each crash, a block of k characters takes $\left(\frac{1}{p} + r\right)\left((1-p)^{-k} - 1\right)$ on average. This is convex in k, so equal blocks are best and only the number of saves remains to choose. That is the contest solution for up to 2000 characters.

9 Chopsticks

Source: 2025 ICPC Asia Taiwan Online Programming Contest, Problem H (QOJ). *Topic:* parity. *Difficulty:* about 1700 (CF scale).

Problem. A box holds 12 chopsticks: 4 red, 4 blue and 4 green. You draw 6 of them uniformly at random and hand them out to 3 guests in pairs, making as many single-colour pairs as possible. Find the expected number of guests who get a mixed pair.

Solution. Key idea: the number of mixed pairs is half the number of colours drawn an odd number of times.

Pair up chopsticks of the same colour as far as possible. Each colour drawn an odd number of times leaves one chopstick over, and the leftovers, all of different colours, form the mixed pairs. So if O colours have an odd count, there are $O/2$ mixed pairs. By linearity and symmetry,

$$\mathbb{E}[\text{mixed pairs}] = \frac{1}{2} \sum_{\text{colours}} \mathbb{P}(\text{odd count}) = \frac{3}{2}\,\mathbb{P}(\text{odd number of reds}).$$

An odd number of reds means 1 or 3 reds, so

$$\mathbb{P}(\text{odd number of reds}) = \frac{\binom{4}{1}\binom{8}{5} + \binom{4}{3}\binom{8}{3}}{\binom{12}{6}} = \frac{224 + 224}{924} = \frac{16}{33},$$

and the expected number of mixed pairs is $\frac{3}{2} \cdot \frac{16}{33} = \frac{8}{11}$. □

Remark. The contest has up to $5 \cdot 10^5$ types. Then one writes $\mathbb{P}(\text{odd}) = \frac{1}{2}\left(1 - \mathbb{E}[(-1)^X]\right)$ and reads $\mathbb{E}[(-1)^X]$ off the coefficient of z^{2n} in $(1-z)^k(1+z)^{s-k}$. With only two colours the trick is not needed, because the number of mixed pairs is just the indicator that the red count is odd; for 8 red and 8 blue with 10 drawn the answer is $\frac{72}{143}$.